

Cells — The Building Blocks of Life

Plant vs Animal cells · Cell organelles and functions · Microscope use

Key Biology Vocabulary — ESL Guide

Cell Theory — The Foundation of Biology

- All living things are made of cells — from a single bacterium to a blue whale (37 trillion cells). Cells are universal to life on Earth.
- The cell is the basic unit of life — the smallest thing that can carry out all life processes: nutrition, respiration, growth, reproduction, response.
- All cells come from pre-existing cells — cells divide to make new cells. No cell arises spontaneously.

Animal Cell Organelles — Functions & Analogies

Plant vs Animal Cells — Three Key Differences

Using a Microscope — Steps & Magnification Formula

Knowledge Check — Questions & Model Answers

- Which THREE structures are found in plant cells but NOT animal cells? → Cell wall, Chloroplasts, Large central vacuole. Each relates to the plant's unique needs: support, photosynthesis, turgidity.
- What is the function of the mitochondria? → Cellular respiration — releasing energy (ATP) from glucose. More active cells (e.g. muscle cells) contain more mitochondria.
- Eyepiece ×10, objective ×40 — total magnification? → $10 \times 40 = \times 400$. The image appears 400 times larger than the actual specimen.
- What material did Robert Hooke observe when he first saw cells in 1665? → CORK. He named the box-like compartments "cells" after the Latin cella (small room).

- Why does a wilting plant suggest its vacuoles are not full? → Turgor pressure: the central vacuole full of water pushes outward against the cell wall, making cells firm. Without water, the vacuole deflates → cell goes limp → plant wilts.